Developing an HLA Curation Framework in ClinGen for Complex Diseases

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Background

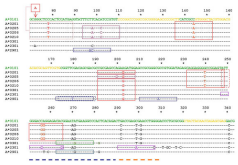
The Human Leukocyte Antigen Expert Panel (HLA EP) under the Rheumatologic Autoimmune Disease Clinical Domain Working Group (RAD CDWG) was established to create an HLA-specific curation framework for ClinGen (https://www.clinicalgenome.org) and to curate the literature for relevant genes and associations for use in precision medicine. The current ClinGen framework was designed for monogenic disorders. Therefore, we are designing a tool to absorb the complex HLA nomenclature and present a framework for curation for complex disease associations.

Goals

- Provide guidance for standardized usage of the WHO Nomenclature in HLA-related complex disease studies to the genomics community
- Establish a team to design an HLA curation framework
- Apply the framework to curate HLA nomenclature
- Identify and recruit HLA experts to participate in gene and variant curation efforts for non-Mendelian disease associations
- Recruit PGx experts to participate in gene and variant curation efforts for drug interactions with HLA molecules
- Generate allele frequency world maps from various databases

HLA Nomenclature

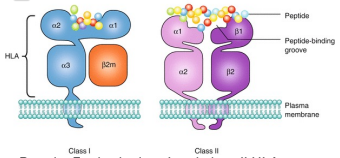
The HLA region on chromosome 6 encodes key components of the adaptive immune system that presents antigens to immune monitoring cells. With 35,000 HLA alleles known, this is the most diverse genomic region.



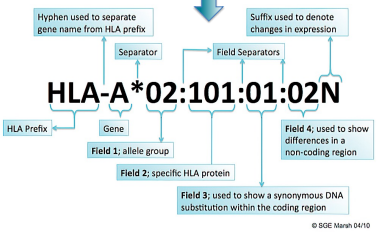
IPD-IMGT/HLA Database

The International ImMunoGeneTics Project
https://www.ebi.ac.uk/ipd/imgt/hla/
Official Repository,
managed by the WHO Nomenclature Committee for Factors of the HLA System, which has managed over 86 releases of HLA allele name and sequence data since 1998

DNA sequence: HLA nucleotide sequences are highly polymorphic and often reflect ancient polymorphism, which makes SNP-based HLA genotyping difficult



Protein: For both class I and class II HLA molecules amino acid diversity is high in the peptide binding groove



Framework Draft

Curators will enter data collected from the literature characterizing associations between HLA alleles and specific diseases. The focus of the framework will be on specific diseases, as opposed to cataloguing associations for a specific allele. Multiple disease-associations will likely be recorded for specific alleles, as the curation framework grows. The ClinGen system will use a series of prompts to guide curators through the necessary fields for curation of a given disease. The ClinGen HLA allele curation framework will assume the curators are not HLA experts. A completed curation will use a point system to characterize the level of evidence for a given association. Multiple studies that identify an association with that specific allele will contribute to a higher (for supporting information) or lower level of evidence (or for conflicting data). In cases where multiple allele associations are included in a specific publication, curators will have the option to modify data already in the ClinGen framework for that publication and modify it accordingly to reduce manual entry burdens.

Input HLA-allele into curation framework

- HLA allele names are validated as they are entered into the framework.
- Converts HLA allele names published under previous IPD-IMGT/HLA Database versions to those in the current database version.
- HLA haplotypes (alleles at different HLA loci determined to be transmitted on the same chromosome), can be linked to each other for curation, but will be stored in the ClinGen Framework as independent alleles.

Disease association

- Curators can select associations from a preexisting disease database populated using Monarch Disease Ontology (MonDO) IDs https://www.ebi.ac.uk/ols/ontologies/mondo

HLA Typing

- Studies describing HLA alleles defined with higher resolution methods will be given greater weight than those describing HLA alleles defined with lower-resolution methods.
For example, Serological data < HLA Imputation < Exon Sequencing < Full Gene Sequence

Sample Size

- Because association confidence is proportional to cohort size, associations derived from larger cohorts will be weighted higher than those derived from smaller cohorts.

Population

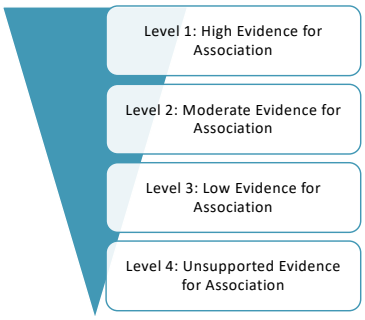
- This field will allow the user to input ancestry or ancestries of the cohort used in the curated paper.
- Population identifiers for curated cohorts will be derived from the Human Ancestry Ontology catalog. For example, the curator can select from various countries, continents, and regions that would be most appropriate for the study (Ancestry > Oceanian > Native Hawaiian) https://www.ebi.ac.uk/ols/ontologies/hancestro

Study Design and Statistical Type

- Curators can select specific study types (e.g., cohort, retrospective, etc.), and the framework will populate data-entry options specific to the statistics applied for that study type. For example, for logistic regression, fields for 'p-value', '95% confidence interval', and 'Odds Ratio' will appear.

Evidence will be weighed based on the input, a score will be computed, and a level will be assigned. A sentence for the allele curation will be generated that will give a clear summary of the association.

Levels of Evidence for Complex Disease Associations



Evidence has been replicated across multiple studies and has a strong level of association between the HLA allele and the disease

Curations report a moderate level of association with the disease. Multiple independent curations are necessary to achieve this Level 2

There is weak evidence for the association with the HLA allele. To go up in the Levels of Association, more curations would be needed to establish higher resolution HLA Typing, larger Sample Sizes or a stronger Study Design

Curation unsupported, generally a sign of conflicting evidence or low level of support for the association

Levels 1 and Level 2 will be split into A and B categories: Level 1A, 1B, 2A, 2B.

A: the disease association for the HLA allele has been replicated in multiple populations
B: the disease association for the HLA allele has been replicated in a single population

Example Output

Level 1A

HLA-B27 has a **high-level association (1A)** for **susceptibility** to **Ankylosing spondylitis**. This association demonstrates an increased risk across **multiple cohorts** and in **multiple populations** worldwide.

Future Directions

We are in the process of conducting an intensive literature review of HLA disease associations to determine : 1) That our framework captures all the information and fields necessary to curate a wide range of HLA disease associations. 2) To define a "scoring" system that accurately reflects the "Levels of Evidence" and determine the appropriate thresholds for each of the framework categories such as HLA Typing, Sample Size, Statistical Type, etc.

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